

CHƯƠNG 6

Exercise 5: Take the following Python code that stores a string: ‘ str = ‘X-DSPAM-Confidence:0.8475’

Use find and string slicing to extract the portion of the string after the colon character and then use the float function to convert the extracted string into a floating point number.

Code:

```
str = 'X-DSPAM-Confidence:0.8475'
```

```
# Tìm vị trí của dấu hai chấm
```

```
vi_tri = str.find(':')
```

```
# Cắt chuỗi từ sau dấu hai chấm
```

```
chuoi_so = str[vi_tri + 1:]
```

```
# Chuyển chuỗi thành số thực
```

```
gia_tri = float(chuoi_so)
```

```
# In ra kết quả
```

```
print(gia_tri)
```

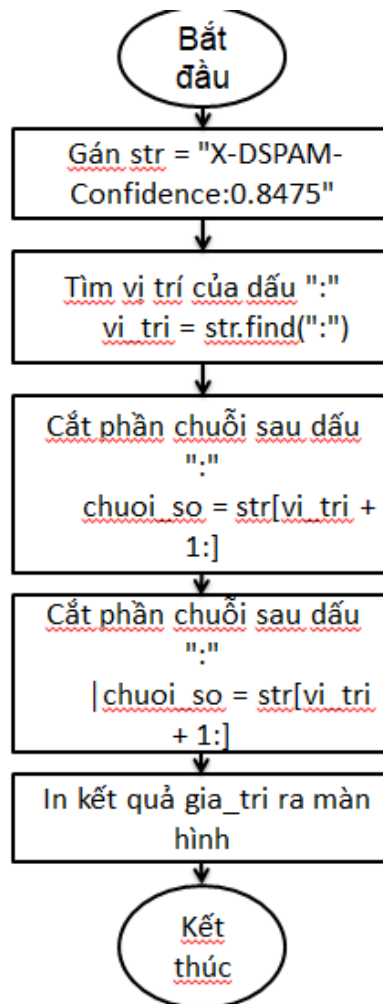
Nhập code:

```
1 str = 'X-DSPAM-Confidence:0.8475'
2
3 # Tìm vị trí của dấu hai chấm
4 vi_tri = str.find(':')
5
6 # Cắt chuỗi từ sau dấu hai chấm
7 chuo_i_so = str[vi_tri + 1:]
8
9 # Chuyển chuỗi thành số thực
10 gia_tri = float(chuo_i_so)
11
12 # In ra kết quả
13 print(gia_tri)
```

Kết quả :

```
PS D:\11> & "C:\Users\ANH TUAN\AppData\Local\Programs\Python\Python313\python.exe" d:/11/hjhj
0.8475
PS D:\11> █
```

Lưu đồ giải thuật :



Exercise 6: Read the documentation of the string methods at <https://docs.python.org/3.5/library/stdtypes.html#string-methods>

You might want to experiment with some of them to make sure you understand how they work. `strip` and `replace` are particularly useful.

The documentation uses a syntax that might be confusing. For example, in `find(sub[, start[, end]])`, the brackets indicate optional arguments. So `sub` is required, but `start` is optional, and if you include `start`, then `end` is optional.

Một số phương thức chuỗi quan trọng và ví dụ minh họa:

— Xóa khoảng trắng ở đầu và cuối chuỗi:

```
text = "  hello world  "
```

```
print(text.strip())
```

Nhập code:

```
1 text = "  hello world  "
2 print(text.strip())      # Kết quả: 'hello world'
3
```

Kết quả:

```
PS D:\11> & "C:\Users\ANH TUAN\AppData\Local\Programs\Python\Python313\python.exe" d:/11/hjhj
hello world
PS D:\11> 
```

— Thay thế ký tự hoặc chuỗi con

```
text = "I like apples"
```

```
print(text.replace("apples", "bananas"))
```

Nhập code:

```
1 text = "I like apples"
2 print(text.replace("apples", "bananas"))
3
```

Kết quả:

```
PS D:\11> & "C:\Users\ANH TUAN\AppData\Local\Programs\Python\Python313\python.exe" d:/11/hjhj
I like bananas
PS D:\11> 
```

— Tìm vị trí đầu tiên của chuỗi con

```
text = "Hello Python"
```

```
print(text.find("Python"))
```

```
print(text.find("Java"))
```

Nhập code:

```
1 text = "Hello Python"
2 print(text.find("Python")) # 6
3 print(text.find("Java"))   # -1 (không tìm thấy)
4
```

Kết quả:

```
PS D:\11> & "C:\Users\ANH TUAN\AppData\Local\Programs\Python\Python313\python.exe" d:/11/hjhj
6
-1
PS D:\11> 
```

— Chuyển chữ thường / hoa

```
text = "PyThOn"
```

```
print(text.lower())
```

```
print(text.upper())
```

Nhập code:

```
1 text = "PyThOn"
2 print(text.lower()) # 'python'
3 print(text.upper()) # 'PYTHON'
4
```

Kết quả:

```
python
PYTHON
PS D:\11> 
```

— Tách chuỗi thành danh sách

```
line = "one,two,three"
```

```
words = line.split(",")
```

```
print(words)
```

Nhập code:

```
1 line = "one,two,three"
2 words = line.split(",")
3 print(words)
4
```

Kết quả:

```
PS D:\11> & "C:\Users\ANH TUAN\AppData\Local\Programs\Python\Python313\python.exe"
['one', 'two', 'three']
PS D:\11> █
```

— Nối danh sách thành chuỗi

```
words = ['one', 'two', 'three']
```

```
result = "-".join(words)
```

```
print(result)
```

Nhập code:

```
1 words = ['one', 'two', 'three']
2 result = "-".join(words)
3 print(result) # 'one-two-three'
4
```

Kết quả:

```
PS D:\11> & "C:\Users\ANH TUAN\AppData\Local\Programs\Python\Python313\python
one-two-three
PS D:\11> █
```

CHƯƠNG 7

Exercise 1: Write a program to read through a file and print the contents of the file (line by line) all in upper case

```
ten_file = input("Nhap ten file: ")

try:
    tep = open(ten_file)
except:
    print("Khong mo duoc file:", ten_file)
    quit()

for dong in tep:
    dong = dong.rstrip()
    print(dong.upper())
```

Nhập code:

```
fname = input("Enter file name: ")

try:
    fhand = open(fname)
except:
    print("File cannot be opened:", fname)
    quit()

for line in fhand:
    line = line.rstrip()
    print(line.upper())
```

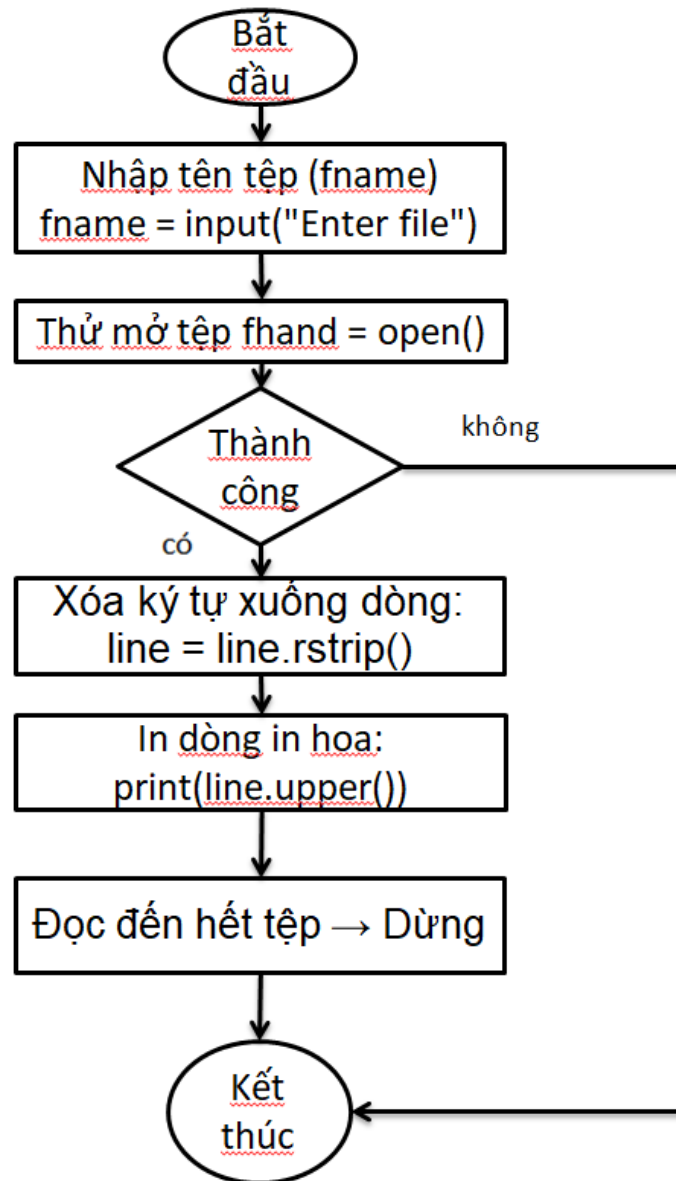
Kết quả:

```
GRADEBOOK/TRUNK/SERVICE/API/SRC/JAVA/ORG/SAKAIPROJECT/SERVICE/GRADEBOOK/SHARED/GRADEBOOKSERVICE.JAVA
GRADEBOOK/TRUNK/SERVICE/IMPL/SRC/JAVA/ORG/SAKAIPROJECT/COMPONENT/GRADEBOOK/GRADEBOOKSERVICEHIBERNATEIMPL.JAVA
LOG:
SAK-12175
HTTP://BUGS.SAKAIPROJECT.ORG/JIRA/BROWSE/SAK-12175
CREATE METHODS REQUIRED FOR GB INTEGRATION WITH THE ASSIGNMENT2 TOOL
GETGRADEDEFINITIONFORSTUDENTFORITEM

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THIS AUTOMATIC NOTIFICATION MESSAGE WAS SENT BY SAKAI COLLAB (HTTPS://COLLAB.SAKAIPROJECT.ORG/PORTAL) FROM THE SOURCE SITE.
YOU CAN MODIFY HOW YOU RECEIVE NOTIFICATIONS AT MY WORKSPACE > PREFERENCES.

FROM ZQIAN@UMICH.EDU FRI JAN 4 10:17:43 2008
RETURN-PATH: <POSTMASTER@COLLAB.SAKAIPROJECT.ORG>
RECEIVED: FROM MURDER (MAIL.UMICH.EDU [141.211.14.97])
        BY FRANKENSTEIN.MAIL.UMICH.EDU (CYRUS V2.3.8) WITH LMTPA;
        FRI, 04 JAN 2008 10:17:43 -0500
X-SIEVE: CMU SIEVE 2.3
RECEIVED: FROM MURDER ([UNIX SOCKET])
        BY MAIL.UMICH.EDU (CYRUS V2.2.12) WITH LMTPA;
        FRI, 04 JAN 2008 10:17:42 -0500
RECEIVED: FROM CREEPSHOW.MR.ITD.UMICH.EDU (CREEPSHOW.MR.ITD.UMICH.EDU [141.211.14.84])
        BY SLEEPERS.MAIL.UMICH.EDU ( ) WITH ESMTP ID M04FHGFS011536;
        FRI, 4 JAN 2008 10:17:42 -0500
```

Lưu đồ giải thuật:



Exercise 2: Write a program to prompt for a file name, and then read through the file and look for lines of the form:

X-DSPAM-Confidence:0.8475

When you encounter a line that starts with “X-DSPAM-Confidence:” pull apart the line to extract the floating-point number on the line. Count these lines and then compute the total of the spam confidence values from these lines. When you reach the end of the file, print out the average spam confidence

```
fname = input("Enter the file name: ")
```

```
try:
```

```
    fhand = open(fname)
```

```
except:
```

```
    print("File cannot be opened:", fname)
```

```
    quit()
```

```
count = 0
```

```
total = 0.0
```

```
for line in fhand:
```

```
    if not line.startswith("X-DSPAM-Confidence:"):

```

```
        continue
```

```
    value = float(line.split(":")[1])
```

```
    total += value
```

```
    count += 1
```

```
if count > 0:
```

```
    print("Average spam confidence:", total / count)
```

```
else:
```

```
print("No X-DSPAM-Confidence lines found.")
```

Nhập code:

```
fname = input("Enter the file name: ")

try:
    fhand = open(fname)
except:
    print("File cannot be opened:", fname)
    quit()

count = 0
total = 0.0

for line in fhand:
    if not line.startswith("X-DSPAM-Confidence:"):
        continue
    value = float(line.split(":")[1])
    total += value
    count += 1

if count > 0:
    print("Average spam confidence:", total / count)
else:
    print("No X-DSPAM-Confidence lines found.")
```

Kết quả:

```
MODIFIED:
GRADEBOOK/BRANCHES/ONCOURSE_2-4-2/APP/UI/SRC/JAVA/ORG/SAKAIPROJECT/TOOL/GRADEBOOK/UI/ROSTERBEAN.JAVA
LOG:
SVN MERGE -C 35014 HTTPS://SOURCE.SAKAIPROJECT.ORG/SVN/GRADEBOOK/TRUNK
U APP/UI/SRC/JAVA/ORG/SAKAIPROJECT/TOOL/GRADEBOOK/UI/ROSTERBEAN.JAVA

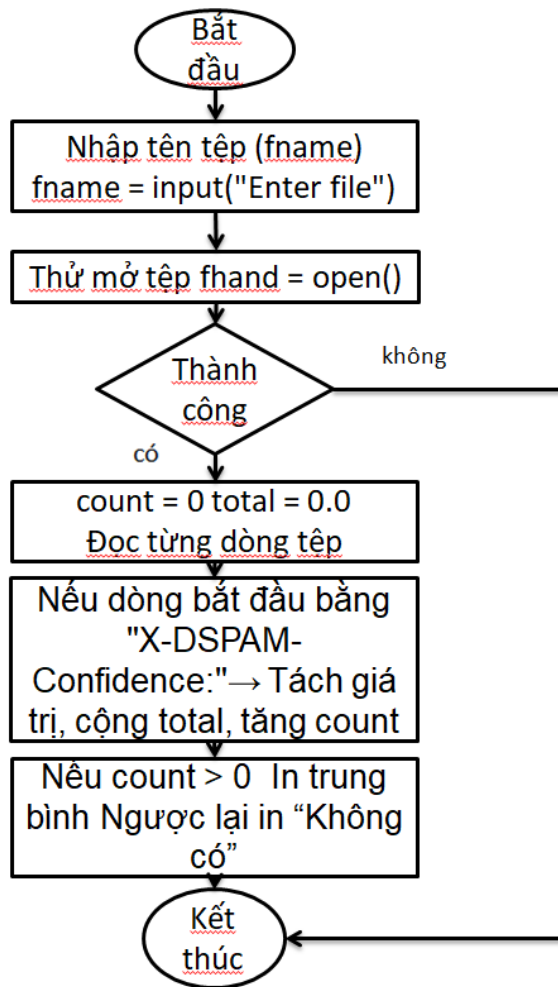
SVN LOG -R 35014 HTTPS://SOURCE.SAKAIPROJECT.ORG/SVN/GRADEBOOK/TRUNK
-----
R35014 | WAGNERMR@IUPUI.EDU | 2007-09-12 16:17:59 -0400 (WED, 12 SEP 2007) | 3 LINES

SAK-11458
HTTP://BUGS.SAKAIPROJECT.ORG/JIRA/BROWSE/SAK-11458
COURSE GRADE DOES NOT APPEAR ON "ALL GRADES" PAGE IF NO CATEGORIES IN GB
-----

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YOU CAN MODIFY HOW YOU RECEIVE NOTIFICATIONS AT MY WORKSPACE > PREFERENCES.

PS C:\Users\bacap\Downloads> c:: cd 'c:\Users\bacap\Downloads'; & 'c:\Users\bacap\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\bacap\.vscode\extensions\ms-python.debugpy-2025.14.1-win32-x64\bundled\libs\debugpy\launcher' '64515' '-.' 'C:\Users\bacap\Downloads\BT2_TrinhAnhTuan.py'
Enter the file name: mbox-short.txt
Average spam confidence: 0.7507185185185187
PS C:\Users\bacap\Downloads>
```

Lưu đồ giải thuật:



Exercise 3: Sometimes when programmers get bored or want to have a bit of fun, they add a harmless Easter Egg to their program. Modify the program that prompts the user for the file name so that it prints a funny message when the user types in the exact file name “na na boo boo”. The program should behave normally for all other files which exist and don’t exist. Here is a sample execution of the program:

```
ten_file = input("Nhap ten file: ")
```

```
if ten_file.lower() == "na na boo boo":
```

```
    print("NA NA BOO BOO TO YOU - You have been punk'd!")
```

```
quit()
```

```
try:
```

```
    tep = open(ten_file)
```

```
except:
```

```
    print("Khong mo duoc file:", ten_file)
```

```
    quit()
```

```
dem = 0
```

```
tong = 0.0
```

```
for dong in tep:
```

```
    if not dong.startswith("X-DSPAM-Confidence:"):

```

```
        continue
```

```
    so = float(dong.split(":")[1])
```

```
    tong += so
```

```
    dem += 1
```

```
if dem > 0:
```

```
    print("Trung binh spam confidence:", tong / dem)
```

```
else:
```

```
print("Khong tim thay dong X-DSPAM-Confidence nao.")
```

Nhập code:

```
ten_file = input("Nhap ten file: ")

if ten_file.lower() == "na na boo boo":
    print("NA NA BOO BOO TO YOU - You have been punk'd!")
    quit()

try:
    tep = open(ten_file)
except:
    print("Khong mo duoc file:", ten_file)
    quit()

dem = 0
tong = 0.0

for dong in tep:
    if not dong.startswith("X-DSPAM-Confidence:"):
        continue
    so = float(dong.split(":")[1])
    tong += so
    dem += 1

if dem > 0:
    print("Trung binh spam confidence:", tong / dem)
else:
    print("Khong tim thay dong X-DSPAM-Confidence nao.")
```

Kết quả:

```
Average spam confidence: 0.7507185185185187
PS C:\Users\bacap\Downloads> mbox-short.txt
mbox-short.txt : The term 'mbox-short.txt' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was
included, verify that the path is correct and try again.
At line:1 char:1
+ mbox-short.txt
~ ~~~~~
+ CategoryInfo          : ObjectNotFound: (mbox-short.txt:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

Suggestion [3,General]: The command mbox-short.txt was not found, but does exist in the current location. Windows PowerShell does not load commands from the current location by default.
If you trust this command, instead type: ".\mbox-short.txt". See "get-help about_Command_Precedence" for more details.
PS C:\Users\bacap\Downloads> c:: cd 'c:\Users\bacap\Downloads'; & 'c:\Users\bacap\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\bacap\.vscode\extensions\ms-python.debugg
v-2025.14.1-win32-x64\bundled\libs\debugpy\launcher' '57827' '--' 'C:\Users\bacap\Downloads\BT3_TrinhAnhTuan.py'
Nhap ten file: mbox-short.txt
Trung binh spam confidence: 0.7507185185185187
PS C:\Users\bacap\Downloads> c:: cd 'c:\Users\bacap\Downloads'; & 'c:\Users\bacap\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\bacap\.vscode\extensions\ms-python.debugg
v-2025.14.1-win32-x64\bundled\libs\debugpy\launcher' '59292' '--' 'C:\Users\bacap\Downloads\BT3_TrinhAnhTuan.py'
Nhap ten file: khongton.txt
Khong mo duoc file: khongton.txt
PS C:\Users\bacap\Downloads> c:: cd 'c:\Users\bacap\Downloads'; & 'c:\Users\bacap\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\bacap\.vscode\extensions\ms-python.debugg
v-2025.14.1-win32-x64\bundled\libs\debugpy\launcher' '61563' '--' 'C:\Users\bacap\Downloads\BT3_TrinhAnhTuan.py'
Nhap ten file: na na boo
NA NA BOO BOO TO YOU - You have been punk'd!
PS C:\Users\bacap\Downloads>
```

Lưu đồ giải thuật:

